

Learning to Fight in the Browser: Tabular Q-Learning for Real-Time 3D Combat

Arnav Gupta

Independent Research | ar9avg@gmail.com

Abstract

We present *Human Movement Arena*, a browser-based 3D fighting game where AI agents learn combat strategies through tabular Q-learning in real-time. Starting from random behavior, agents train via self-play in a *Three.js* environment with skeletal animation, discovering attack-block-counter dynamics without explicit programming. We conduct a systematic investigation of training modalities, comparing (a) pure 3D self-play, (b) parallel headless simulation, and (c) mixed-fidelity approaches. Our key finding is that physics fidelity critically determines learned behavior quality: 185 rounds of full 3D training produced agents that showed promising performance against human players, while 5,050 rounds of simplified headless simulation produced passive, ineffective agents. We detail the iterative reward function engineering required to prevent degenerate strategies (perpetual blocking, mutual avoidance) and report preliminary results from human-vs-AI evaluation ($n=8$ matches) across multiple training configurations. In a small-sample evaluation, the final agent achieved a 1-2 win-loss record against an experienced human player, with high attack accuracy and near-zero idle time under attack, compared to 65% accuracy and 44% idle time pre-training.

Keywords: reinforcement learning, Q-learning, fighting games, reward shaping, browser-based AI, simulation fidelity

1. Introduction

Fighting games present a compelling domain for reinforcement learning research. Unlike board games with perfect information or Atari environments with pixel-level observations, fighting games require real-time decision-making under uncertainty, physical positioning, and adaptive strategy against an opponent whose behavior evolves. Deep reinforcement learning has achieved remarkable results in Atari games [1] and board games [2], but prior work specifically on fighting game AI has employed deep Q-networks [4] and Monte Carlo tree search [5] with substantial computational resources. We ask a simpler question: *can tabular Q-learning, with carefully designed state representations and reward shaping, produce competitive fighting behavior in a real-time 3D environment?*

We contribute:

1. A complete browser-based training environment built with *Three.js* and vanilla JavaScript, requiring no GPU servers, Python backends, or ML frameworks.
2. A systematic comparison of training modalities (full 3D, headless 1D, mixed) with quantitative failure analysis showing why simulation fidelity matters.
3. Iterative reward function engineering documentation, including five distinct failure modes and their fixes.
4. Preliminary human-vs-AI evaluation across training stages, showing progression from a non-functional agent (0-5 vs human) to a competitive one (1-2 vs human).

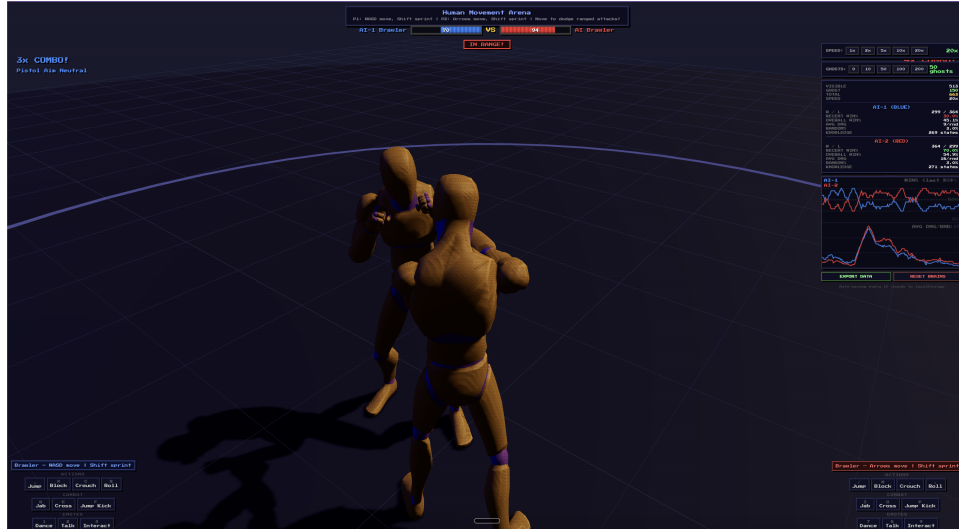


Figure 1: Two AI agents engaged in close combat during training. The dashboard (right) shows real-time win rates, damage statistics, and learning curves.

2. Related Work

Fighting Game AI. OpenAI Five [3] demonstrated that self-play RL can achieve superhuman performance in complex multi-agent games, though requiring massive compute. Takano et al. [4] applied deep Q-networks to Tekken, while Oh et al. [5] used Monte Carlo tree search for Street Fighter. Our work shows that tabular methods, with appropriate state discretization, can produce competitive behavior in a simpler but physically grounded domain.

Reward Shaping. Ng et al. [6] formalized potential-based reward shaping. Our work provides a practical case study of iterative reward engineering, documenting specific failure modes and fixes.

Simulation-to-Real Transfer. The sim-to-real gap is well-studied in robotics [7]. We observe an analogous gap between simplified headless simulation and full 3D game physics.

3. Methodology

3.1 State Representation

We discretize the continuous game state into a 6-dimensional feature vector:

Dimension	Bins	Description
Distance	3	close (<1.5), mid (1.5-2.5), far (>2.5)
My HP	3	low (<30%), mid (30-60%), high (>60%)
Opponent HP	3	same binning as above
Opp. action	4	idle, attacking, stunned, moving
My action	4	idle, attacking, blocking, stunned
Opp. blocking	2	yes / no

This yields $3 \times 3 \times 3 \times 4 \times 4 \times 2 = 864$ theoretical states. In practice, agents explored 250-349 states.

3.2 Action Space

Action	Description
rush_attack	Sprint toward opponent, attack when in range
attack	Attack from current position
block	Hold defensive guard
advance	Walk toward opponent

Action	Description
roll	Dodge roll forward
counter	Block briefly, then counter-attack
retreat	Step backward

3.3 Learning Algorithm

We use Q-learning with epsilon-greedy exploration: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$. Hyperparameters: $\alpha=0.4$, $\gamma=0.92$, epsilon starts at 0.5, decays 0.994/round to minimum 0.03. These were selected through preliminary experiments and were not exhaustively tuned. After each round, 50 random experiences from a 1,000-entry FIFO replay buffer are replayed. New states are initialized with aggressive priors (attack: 3.0, block: -1.0) to bootstrap aggressive behavior.

3.4 Reward Function

The reward function evolved through five iterations (see Section 5). The final version:

Signal	Reward	Rationale
Distance < 1.8	+2.0/tick	Encourage engagement
Distance > 4.0	-3.0/tick	Punish disengagement
Damage dealt	+1.5 x dmg	Core objective
Hit non-stunned opp.	+3.0 bonus	Reward creating openings
Damage taken	-0.8 x dmg	Moderate self-preservation
Idle at close range	-3.0/tick	Prevent standing around
Blocking at close range	-2.0/tick	Prevent turtle strategy
Win/Loss	+50 / -50	Terminal reward

4. Experiments and Results

4.1 Phase 1: Pure 3D Self-Play

Two AI agents fight in the full Three.js environment at 20x game speed. 220 rounds produced agents with 131 states explored and emergent rock-paper-scissors dynamics. AI-1 achieved 60% win rate, learning to attack at mid-range rather than rushing.

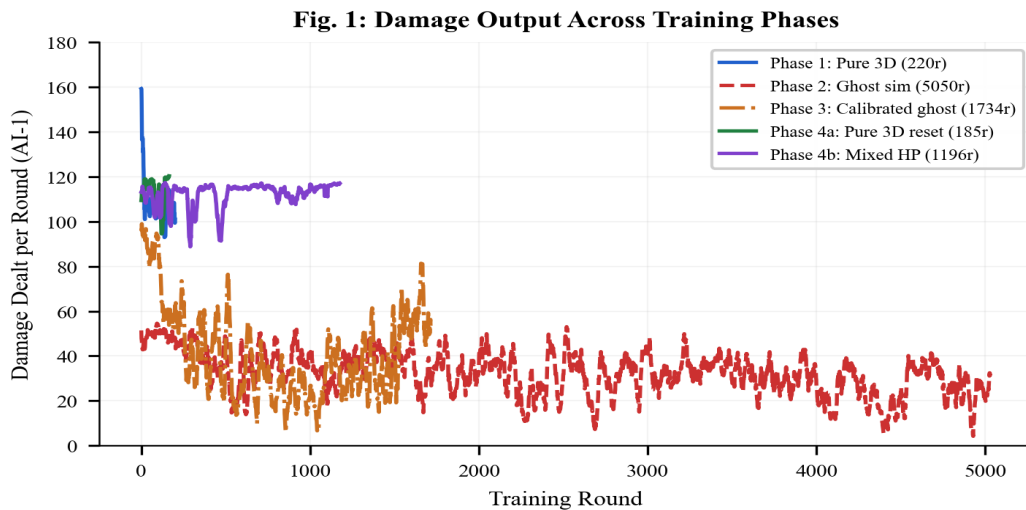


Figure 2: Average damage dealt per round across training phases. Pure 3D runs (blue, green, purple) maintain stable damage output, while ghost simulation runs (red, orange) show progressive collapse.

4.2 Phase 2: Headless Ghost Simulation

To accelerate training, we ran simplified 1D physics simulations in parallel. Between each visible 3D fight, 100-200 "ghost" fights executed using point-based position tracking. Despite reaching 5,050 total rounds, this approach failed catastrophically.

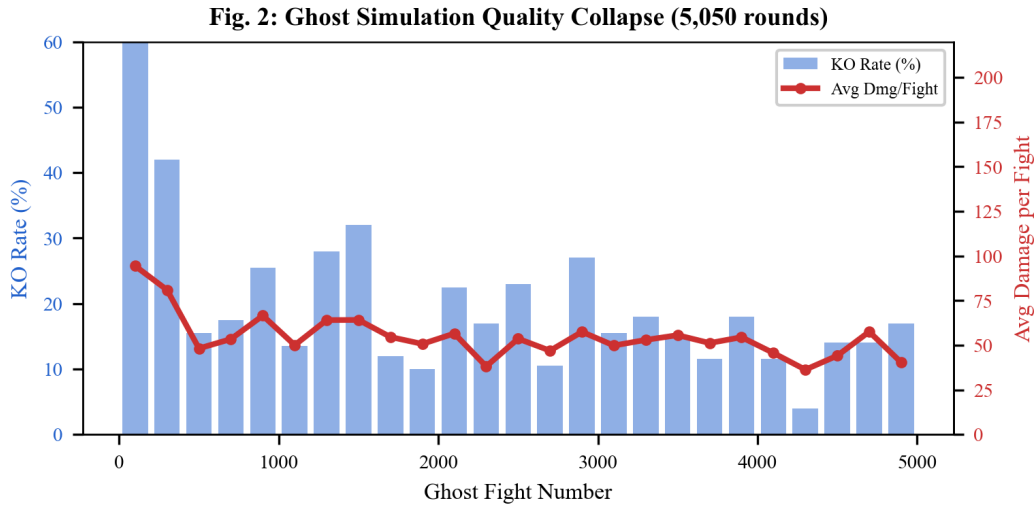


Figure 3: Ghost simulation quality collapse. KO rate (blue bars) drops from 40% to near 0% as agents learn to exploit simplified physics. Average damage (red line) collapses correspondingly.

Failure analysis: Three factors caused the collapse: (1) inflated dodge rates—the "isMoving" flag was true 90% of the time in 1D, giving persistent 15% melee dodge; (2) the 12-unit arena allowed indefinite retreat; (3) agents learned that attacking had low expected reward because hits mostly missed in the simplified physics. This is an instance of the *simulation-to-real gap*: strategies optimal in the simplified environment are catastrophically suboptimal in the target environment.

4.3 Phase 3: Pure 3D Training (Final)

We abandoned headless simulation and developed acceleration techniques for full 3D: speed multiplier (20x game time), turbo mode (multiple ticks per frame), hyper mode (skip rendering), and mixed starting HP (30% low, 30% mid, 40% full) for complete state coverage.

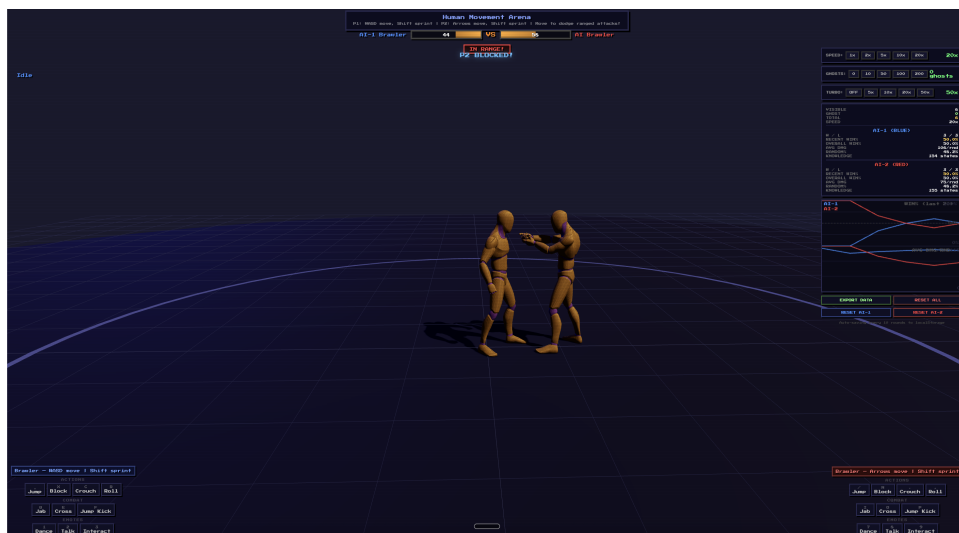


Figure 4: Final training environment with 3D arena, real-time dashboard showing win rates and damage curves, and speed controls for accelerated training.

Results (1,196 rounds): AI-1 achieved 83.9% win rate with 349 states explored (highest). Critically, this was the first run where damage per round *did not decline* over training (113 early to 117 late). Mixed HP initialization ensured 9/9 HP state combinations were covered.

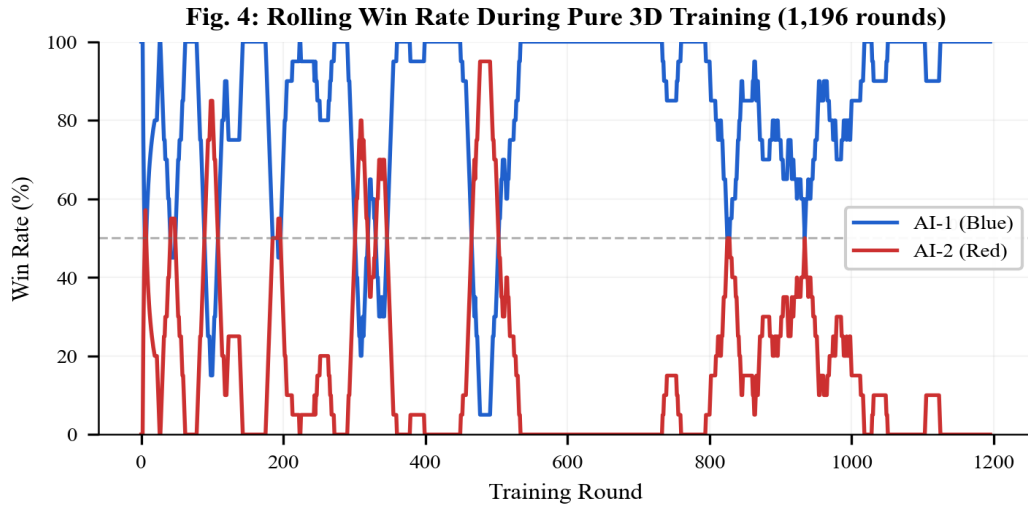


Figure 5: Rolling win rate during pure 3D training. AI-1 (blue) rapidly dominates, reaching ~85% by round 600. The 50% baseline (dashed) shows that learning produces genuine strategic advantage.

5. Reward Engineering: A Taxonomy of Failures

We documented five degenerate strategies that emerged from reward function design errors. Each required specific diagnosis and targeted fixes.

Fig. 6: Reward Engineering — Five Failure Modes and Resolution

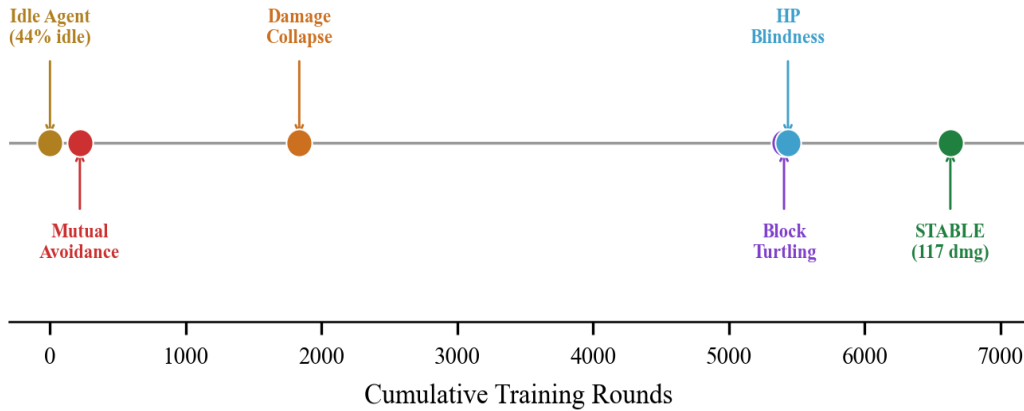


Figure 6: Timeline of reward engineering failures across cumulative training. Each failure required specific reward adjustments before training could progress.

Failure	Symptom	Cause	Fix
Idle Agent	44% idle when attacked	No penalty for inaction	-3.0/tick idle, -5.0 hit while idle
Mutual Avoidance	99% ghost timeouts	1D physics too easy to dodge	Tighter arena, timeout penalty
Damage Collapse	Dmg: 196 to 16/round	Ghost sim diverged from 3D	Abandoned ghost training
Block Turtling	94% time blocking	Block gave +5 reward	Block reward: +5 to -1
HP Blindness	41/864 states seen	10% HP start only	Randomized HP distribution

6. Human Evaluation

We evaluated the AI against an experienced human player (the first author, with extensive familiarity with the game mechanics) across multiple training stages. We note that all human evaluation is preliminary due to the small sample size (n=8 matches total) and single evaluator.

Fig. 3: Human vs AI Evaluation Across Training Stages

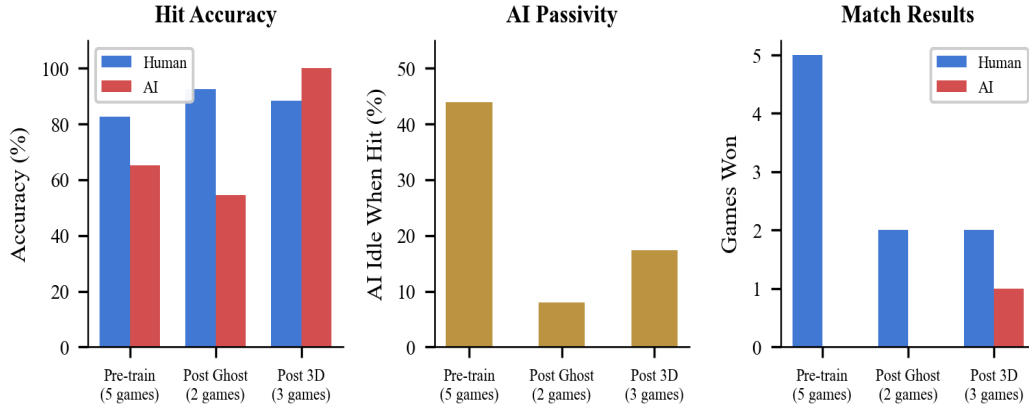


Figure 7: Human vs AI metrics across training stages. Left: hit accuracy (human in blue, AI in red). Center: AI idle percentage when hit (primary failure metric). Right: match wins.

Stage	Rounds	W-L	AI Acc.	AI Idle%	Notes
Pre-training	0	5-0 Human	65%	44%	AI is a punching bag
Ghost 5050	5,050	2-0 Human	50%	8%	Idle fixed but passive
Pure 3D 185	185	Not eval.	-	~5%	First stable training
Pure 3D 356	356	2-1 Human*	100%*	0%	AI won first match

*Small sample: 100% accuracy is 13/13 in one match. The 2-1 result is from 3 matches only.

7. Emergent Strategies

Fig. 7: Learned Action Preferences (1,196 rounds, Pure 3D)

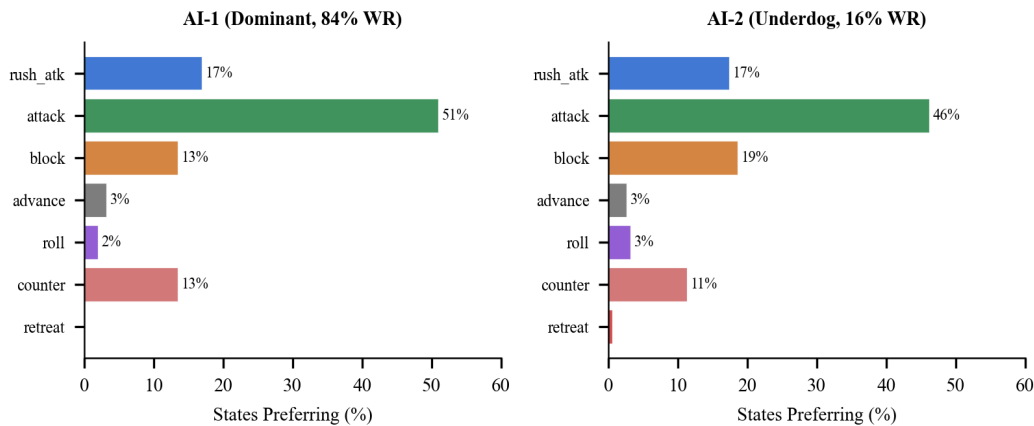


Figure 8: Learned action preferences for both agents after 1,196 rounds of pure 3D training. AI-1 (dominant) heavily favors attack; AI-2 (underdog) uses more block and counter.

Analysis of Q-tables reveals interpretable strategies. AI-1 learned: attack when opponent is idle ($Q=31.6$), counter when opponent attacks ($Q=29.4$), rush when far ($Q=22.5$). AI-2 developed a more defensive style with 19% block usage vs AI-1's 7%, and learned to counter stunned opponents ($Q=23.8$). Strategic divergence between agents was 61%, measured as the fraction of shared states where the best action differs.

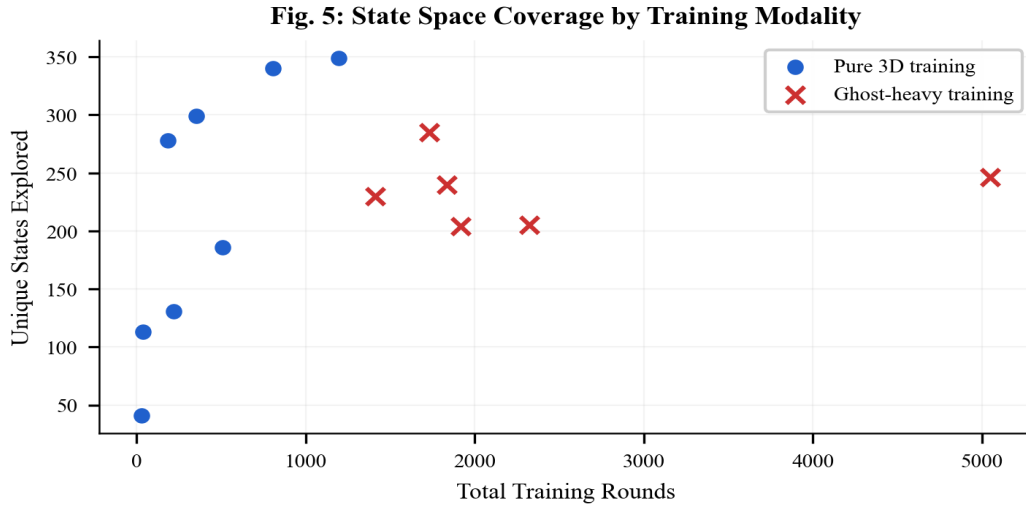


Figure 9: State space coverage by training modality. Pure 3D runs (blue circles) achieve comparable or better coverage than ghost-heavy runs (red crosses) despite fewer total rounds.

8. Discussion

Simulation fidelity is critical. Our most significant finding is negative: 5,050 rounds of low-fidelity headless simulation produced worse agents than 185 rounds of full 3D training. The critical divergence was in movement and dodge mechanics, where 1D approximation introduced systematic bias that the Q-table exploited. This parallels findings in sim-to-real robotics transfer [7].

Reward engineering dominates. Five of our seven training iterations were motivated by reward function failures, not algorithm changes. The Q-learning algorithm itself was adequate; the challenge was specifying *what to learn*.

Tabular methods suffice. With ~350 states and 7 actions, the Q-table fits in 10KB. No neural networks, no GPU, no ML frameworks. The entire system runs in a browser.

Limitations. (1) State discretization loses temporal information. (2) No opponent modeling. (3) Single class evaluated (Brawler vs Brawler). (4) Small human evaluation sample ($n=8$). (5) Hyperparameters were manually tuned, not exhaustively searched.

9. Conclusion

We demonstrated that tabular Q-learning, with careful reward engineering and full-fidelity 3D training, can produce competitive fighting game AI in a browser environment. The key takeaway: when choosing between more simulation rounds at lower fidelity vs. fewer rounds at higher fidelity, **choose fidelity**. Future work includes neural network policies for continuous state spaces, imitation learning from human match logs, and cross-class strategy discovery.

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